LOCALISING A LOCALISED MODULE TWICE GIVES THE LOCALISATION AT THE PRODUCT SET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted proof ends with the author's sentence that well-definedness is easily checked and that showing the two maps are inverse is straightforward; the maps themselves are written out in full and the sentence is kept verbatim inside the span.

Editorial note. The counted spans contain no \ref and no \cite, so the card carries no navigation stub and no bibliography.

Allowed standard prerequisites (editorial).

- The product SS' as a multiplicative subset. Used in the statement and in the displayed definition of the two maps. Supplies the target of the isomorphism and the denominator form ss' used by both maps.
- Absence of a universal property for module localisation at this point. Noted by the author at the start of the counted proof (lines 1525-1527). Explains why the isomorphism must be constructed on elements rather than deduced from a universal property.

Proposition 0.1. *View $S'^{-1}M$ as an A-module, then $S^{-1}(S'^{-1}M)$ is isomorphic to $(SS')^{-1}M$.*

Proof. Note that given a A-module M, we have not proved any universal property for $S^{-1}M$. Hence we cannot reason as in the preceding proof; we have to construct the isomorphism explicitly.

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We define the maps as follows

$$f : S^{-1}(S'^{-1}M) \longrightarrow (SS')^{-1}M, \quad \frac{x/s'}{s} \mapsto x/ss'$$

$$g : (SS')^{-1}M \longrightarrow S^{-1}(S'^{-1}M), \quad x/t \mapsto \frac{x/s'}{s} \text{ for some } s \in S, s' \in S', \text{ and } t = ss'$$

We have to check that these homomorphisms are well-defined, that is, independent the choice of the fraction. This is easily checked and it is also straightforward to show that they are inverse to each other. $\square$